A $10 Chip Could Put a Lab on Your Wrist

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Imagine checking your blood sugar with no needle, just a **sensor** on your wrist. Researchers at the University of Cambridge and a start-up called GlitterinTech built a chip that could one day make that possible. It costs just $10.

The chip reads light to learn what a **substance** is made of. Tools like this are called **spectrometers**. They are used in medicine, factories, food testing and nature studies. Big **spectrometers** work well, but making them small usually makes them less exact. This chip is only a few centimeters wide, yet the team says it is as **precise** as much bigger lab machines.

In tests, the chip correctly named plastics, medicines, coffee, flour and tea every time. It also measured health markers such as blood **glucose**, without needles, in one test over a long period. The researchers say much more testing is needed before it can become a real medical product.

The chip kept working from -20°C to 80°C, so it could suit **wearables**, factories and outdoor **sensors**.

Word Treasure

sensor -- A sensor is a device that notices and measures something.

spectrometers -- Spectrometers are tools that read light to learn what a substance is made of.

substance -- A substance is a kind of material.

precise -- Precise means exact and careful.

glucose -- Glucose is a sugar found in blood.

wearables -- Wearables are devices that people wear on their bodies.

Background you need

A spectrometer is a tool that reads light to learn what something is made of.

Big versions can be hard to make small without losing exactness, but this chip stays small and precise.

It could one day help people check health markers from a wrist sensor, after much more testing.

Break it down

WHO: Researchers at the University of Cambridge and the start-up GlitterinTech

WHAT: They built a small, low-cost chip that reads light to identify substances and measure health markers

WHERE: In tests, with possible future use on wrists, in factories and outdoors

WHY: To make a small spectrometer that stays as precise as much bigger lab machines

Why it matters

If it passes more tests, a wrist sensor could one day let kids check health markers without a needle.

Test what you remember

1. Who built the new chip?

- (A) Doctors at one hospital
- (B) Researchers at the University of Cambridge and GlitterinTech
- (C) Workers at a watch factory
- (D) A team of chefs

2. How does the chip learn what a substance is made of?

- (A) It tastes the substance
- (B) It listens for sounds
- (C) It reads light
- (D) It weighs the substance

3. What usually happens when big spectrometers are made small?

- (A) They get much heavier
- (B) They stop working in factories
- (C) They turn into watches
- (D) They become less exact

4. Which group of items did the chip name correctly every time in tests?

- (A) Milk, eggs, rice and apples
- (B) Rocks, leaves, sand and shells
- (C) Paper, wood, glass and soap
- (D) Plastics, medicines, coffee, flour and tea

5. What did the chip measure without needles in one long test?

- (A) How fast a heart beats
- (B) How much water a person drank
- (C) Body temperature
- (D) Health markers such as blood glucose

6. What do the researchers say must happen before the chip can become a real medical product?

- (A) It must be worn all day
- (B) It must be made larger
- (C) Much more testing is needed
- (D) It must be sold in stores

Answer key (try the quiz first!): 1: B 2: C 3: D 4: D 5: D 6: C

Questions to think about

1. The researchers: They say the chip is as precise as much bigger lab machines, but much more testing is needed before it can become a real medical product.

MY THOUGHTS (at least 20 words):